

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

May 2012

EC 309 Digital Communication

Date: 08.05.2012, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Compare the Digital Carrier system: ASK, FSK and PSK [05]
(b) Define Bit rate and Baud rate. [02]
- Q - 2 (a) Construct a systematic (7, 4) cyclic code and find out the code vector for the data vector 1010. [06]
(b) For a (6,3) code, the generator matrix G is [08]

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

For all eight possible data words, find the corresponding code words and verify that how many errors this code can correct?

OR

- Q - 2 (a) Give the comparison between coded and uncoded systems. [06]
(b) For (6,3) systematic LBC the three parity check digits C_4 , C_5 , and C_6 are [08]
- $C_4 = d_1 + d_2 + d_3$
 $C_5 = d_1 + d_2$
 $C_6 = d_1 + d_3$
- (i) Construct the appropriate generator matrix for this code.
 - (ii) Construct the code generated by this matrix.
 - (iii) Determine the error correcting capabilities of this code.

Q - 3 (a) A zero memory source emits messages m_1 and m_2 with probabilities 0.9 and 0.1 respectively. Find the optimum (Huffman) binary code for this source as well as for its second order extensions. Determine the code efficiencies in each case. [06]

(b) A signal amplitude x is a random variable uniformly distributed in the range $(-1, 1)$. This signal is passed through an amplifier of gain 2. The output y is also a random variable, uniformly distributed in the range $(-2, 2)$ the differential entropy $H(x)$ and $H(y)$. [04]

Consider

$$P(x) = \begin{cases} \frac{1}{2} & |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$P(y) = \begin{cases} \frac{1}{4} & |y| < 2 \\ 0 & \text{otherwise} \end{cases}$$

(c) Explain BPSK with necessary waveforms. [04]

OR

Q - 3 (a) Show that entropy is maximum when all messages are equiprobable. [06]

(b) A source emits messages m_1 and m_2 with probabilities 0.5 and 0.7. Which messages contain more information? Why? [04]

(c) Explain QPSK with necessary wave forms. [04]

SECTION – II

Q - 4 (a) Define CDF and PDF. State the properties of CDF. [05]

(b) Define conditional probabilities. [02]

Q - 5 (a) What are the desirable properties of line coding? Also give comparative study of different line codes. [07]

(b) Draw the block diagram of digital communication system and explain the three major signal processing tasks. List the advantages of Digital Communication over Analog Communication. [07]

OR

Q - 5 (a) Explain briefly Detection Probability. [07]

(b) Write a short note on Scrambling. [07]

Q - 6 (a) Write short note on Central – Limit theorem. [07]

(b) Compare Delta Modulation and Adaptive Delta Modulation. [07]

OR

Q - 6 (a) A binary Symmetric channel (BSC) error probability is p_e . The probability of transmitting 1 is Q , and that of transmitting 0 is $1-Q$. Determine the probabilities of receiving 1 and 0 at the receiver. [07]

(b) State the advantages and disadvantages of DPCM and ADPCM. [07]